



Development and characterization of chemically stabilized ionic liquid membranes-Part I: Nanoporous ceramic supports

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ABSTRACT

Stabilization of a room temperature methylimidazolium cation based ionic liquid (IL) into the pores of ceramic nanofiltration (NF) membranes was achieved by means of a novel in situ, pressure assisted imbibition/reaction method. The complete coverage of the nanopores void space by the synthesized IL phase, introduced an additional parameter that was pursued in parallel to the challenge of stabilization. The purpose was to develop hybrid membranes that will exhibit solely the gas solvation/diffusion properties of the IL. A silane functionalized ionic liquid phase of 1-methyl-3-(1-trialkoxysilylalkyl) imidazolium as cation and hexafluorophosphate as anion, was first synthesized and subsequently grafted onto the pore surface. The CO₂/CO separation performance of the modified membranes was quite significant. The maximum CO₂/CO separation factors achieved were 11 at 30 °C, 25 at 50 °C, 12 at 100 °C and 5 at 230 °C. The respective CO₂ permeability values were 226, 520, 715 and 2000 Barrer. The stability of the developed membranes has been verified, in 3 sequential cycles of heating/cooling, in the temperature range between 25 and 250 °C and under a differential pressure up to 0.5 MPa.

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1. Introduction

Room temperature ionic liquid membranes or else supported ionic liquid membranes (SILMs) have lately gained significant research attention and tend to replace traditional immobilized liquid membranes (ILMs) [1–3] in gas separation applications. The primary reason for the current popularity of SILMs is the negligible vapor pressure of the room temperature ionic liquids (RTILs), a property that ensures no mass loss due to volatilization under the conditions of elevated temperatures and continuously flowing streams usually involved in membrane separation technology. Other important properties of the RTILs that could be exploited are their significant thermal stability, the high diffusion coefficients and above all, their specific solvation capacity for several gaseous solutes of industrial interest. The ability to synthesize application-specific RTILs via chemical modification of the ions, leads to new opportunities for the development of highly selective and, at the same time, highly permeable membranes. Several studies have already confirmed the specific interaction of the ionic liquid anion with the gaseous molecules. Using ATR-IR spectroscopy, Kazarian et al. [4] have found evidence of a weak Lewis acid–base interaction between the CO₂ and the anions of the examined ILs. Furthermore

MD simulations indicated that CO₂ organizes strongly about the IL anion in a “tangent like” configuration that maximizes favorable interactions [5]. The gas absorption capacity of imidazolium based ionic liquids has been extensively studied by several research groups [6–9]. In all these studies, the ILs solvation capacity for CO₂ outperformed this for other gaseous molecules. In this regard, imidazolium based ionic liquids may be established as excellent candidates for gas separations of high industrial interest such as CO₂/H₂, CO₂/N₂, CO₂/CH₄ and CO₂/CO. The first attempts to conjoin membrane “gas separation” technology with ILs appeared in the middle of the past decade with the work of Scovazzo et al. [10]. Since then, much of the related to the ILs research effort is devoted to the development of supported ionic liquid membranes (SILMs) and several ionic liquids and gas mixtures of industrial interest have been studied [11–22]. The separation performance results were quite significant and confirmed the potentiality for future SILMs application. In most of the aforementioned studies, hydrophobic or hydrophilic macroporous filters (0.1–0.45 μm), made of polymers such as polyvinylidene fluoride, polyamides, polytetrafluoroethylene, polysulfone and polyethersulfone, were used as supports, in order to ensure good wetting and facilitate the IL incorporation into the pore structure. The involvement of nanofiltration (pore size < 1 nm), polymeric CF thin film supports (Sterlitech Corporation-YMHLS1905) is for the first time mentioned in 2006 [23,24] and quite attractive results, at least as regards the stability of the developed SILMs, were reported. However, the mechanisms of

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imbibition and stabilization into the nanosize pores were rather unclear. In this context, the authors postulated that the ionic liquid impregnation was due to the good wettability of the polymeric support by the IL phase and have attributed the significant SILMs stability to the strong cationic/anionic interactions with the functional groups existing at the nanopore walls within a radial dimension of 0.1–1 nm.

In addition to the imbibition/stabilization mechanism, there are still some important open issues in the up to date conducted research. Because of temperature limitations imposed by the application of the polymeric supports, the performance of the SILMs at temperatures above 150 °C is rarely evaluated. In one of the recent studies [15], the use of a cross-linked porous nylon-66 support permitted the investigation of the CO₂/H₂ separation capacity at high temperatures up to 300 °C. The results were very promising and the maximum CO₂/H₂ separation factor achieved was 20 at 85 °C. Despite the good separation capacity, the tests were performed at zero differential pressure across the membrane sides and the reproducibility of performance following repeated cycles of heating/cooling was not reported [15]. A more recent study concerning ceramic nanofiltration membranes was devoted to the investigation of the imbibition kinetics of the ionic liquid 1-butyl-3-methylimidazolium-bis(trifluoromethanesulfonyl)imide—[BMIM][Tf2N] inside the pores of the membrane but no further information was provided as regards the gas separation performance of the produced materials [20].

The current study introduces several important novelties concerning the development of SILMs systems. Primarily, the gas separation performance of IL modified ceramic nanofiltration membranes of asymmetric structure (pore size 1–5 nm) is reported herein for the first time. The tests were conducted at elevated temperatures, just below the decomposition temperature of the incorporated ionic liquid phase (usually above 300 °C), under sequential cycles of heating/cooling and at elevated trans-membrane pressures. The in situ method proposed for the development of the SILM system circumvents the drawbacks commonly encountered for ex situ methods when employing oxygen or humidity sensitive liquids. A functionalized (silylated) ionic liquid was first synthesized and then grafted onto the nanopore surface by means of a type II SN2 reaction mechanism between the solid surface hydroxyls and the alkoxy groups of the silylated IL. Thus, the stability of the membrane is for the first time achieved as the result of chemical bonding rather than because of capillary forces or cationic/anionic interactions. Furthermore, aspects concerning the effects of rehydroxylation pretreatments applied to the NF membranes, as well as issues relevant to the mechanism of diffusion in the developed SILMs were investigated. For the later purpose we have also studied the absorption capacity and the absorption kinetics of the synthesized ionic liquid and its silylated analogue. The developed membranes have been proved stable for three sequential cycles of heating/cooling and the CO₂/CO separation factor as a function of temperature exhibited a maximum. The maximum CO₂/CO separation achieved was 25 at 50 °C.

2. Experimental

2.1. Chemicals and materials

Solvents were used as purchased without 'further drying'. Toluene (puriss, Reag ACS, Aldrich), dichloromethane (GC 99.5%, Aldrich), chloroform (GC, Aldrich), methanol (P99.8, Aldrich). Reaction reagents were used as purchased without 'further purification'. Chlorobutane (P99%, Fluka), 1-methylimidazole (P99% redistilled, Aldrich), (chloromethyl)-trimethoxysilane (96%, Aldrich),

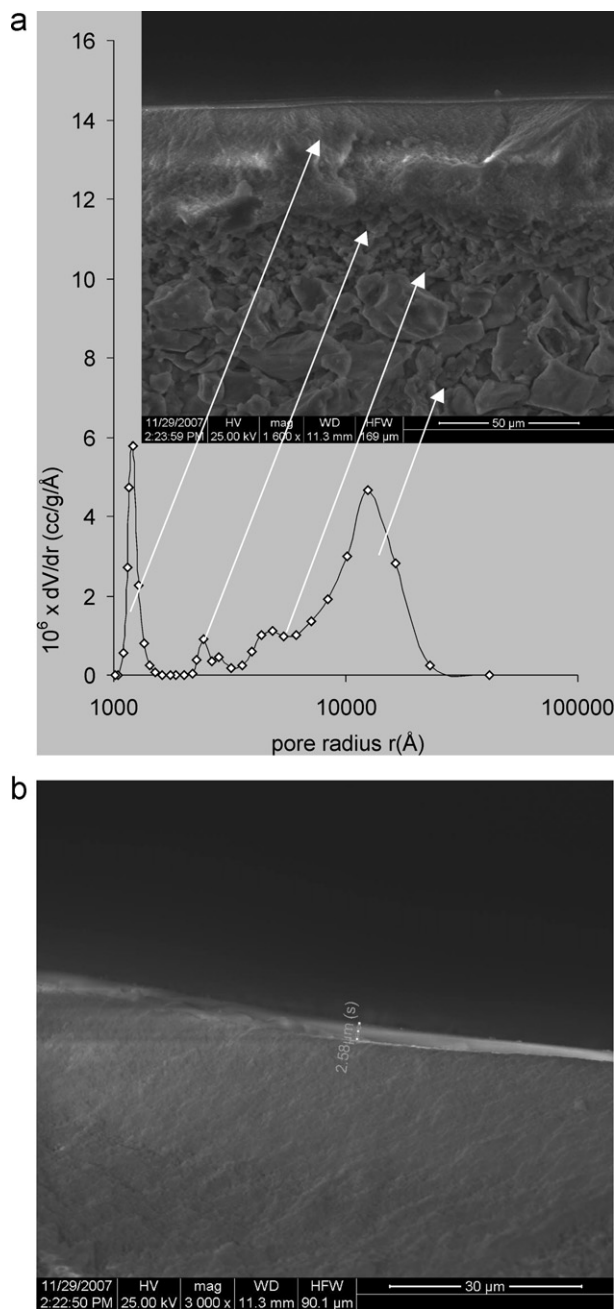


Fig. 1. (a) SEM image of a vertical cross section of the membrane showing the sequential layers of different pore size as defined by mercury porosimetry and, (b) SEM image focusing in the area of the top nanofiltration layer.

(3-chloropropyl)-trimethoxysilane (97%, Aldrich), sodium hexafluorophosphate (98%, Aldrich).

Tubular nanofiltration membranes (tubes of 15 cm length, 0.7 cm ID, 1 cm OD, glazed ends 1.5 cm) with an active NF area of 26 cm² were obtained from Inopor®. The tubes consisted of a macroporous α-alumina support (2.5 μm mean pore size), two intermediate γ-alumina layers of mean pore size 0.5 and 0.25 μm and thickness of 25 and 40 μm, respectively (Fig. 1a) and a top separating layer of about 1.5 μm thickness, located on the inner side of the tubes (Fig. 1b). The pore size for the NF gamma-alumina layer was 5 nm and for the NF silica layer 1 nm. As indicated by the pore size distribution, derived from the mercury intrusion experiments, at the interface between the macroporous alpha-alumina support and the first gamma-alumina layer there is an additional

Table 1

The code of the modified and unmodified membranes. Included is the chemical formula of the grafted ionic liquid.

Sample code	Pore size	Grafted molecule
M1	1 nm	Untreated
SILM1A	–/–	$C_8H_{17}N_2SiO_3PF_6$
SILM1B	–/–	–/–
SILM1C	–/–	$C_{10}H_{21}N_2SiO_3PF_6$
M5	5 nm	Untreated
SILM5A	–/–	$C_8H_{17}N_2SiO_3PF_6$
SILM5B	–/–	$C_{10}H_{21}N_2SiO_3PF_6$

class of pore size resulting from the embedment of the smaller gamma-alumina particles into the void space between the larger alpha-alumina ones that constitute the support. The pore structure characteristics of the nanofiltration layer were elucidated with a combination of single-phase He permeability with LN_2 porosimetry experiments.

In order to test the reproducibility of the proposed IL grafting method and the effect of the silanes alkyl chain length on the imbibition efficiency, we have proceeded with the modification of nanofiltration membranes with different pore size. Thereby two silica (1 nm pore size) and one gamma-alumina (5 nm pore size) membranes were modified with 1-methyl-3-(1-trimethoxysilylmethyl) imidazolium hexafluorophosphate ($C_8H_{17}N_2SiO_3PF_6$) and one silica and one gamma-alumina with 1-methyl-3-(3-trimethoxysilylpropyl) imidazolium hexafluorophosphate ($C_{10}H_{21}N_2SiO_3PF_6$). The sample codes are included in Table 1.

The synthesis of [bmim][PF₆[−]] and 1-methyl-3-(3-trimethoxysilylpropyl) imidazolium hexafluorophosphate, was based on published data with minor changes in what concerns the amount of the synthesized molecules [25,26].

For the preparation of the silylated IL 1-methyl-3-(1-trimethoxysilylmethyl)imidazolium hexafluorophosphate ([smmim][PF₆[−]]) the following procedure was followed:

A mixture of chloromethyl-trimethoxysilane (2 ml, 13.27 mmol) and 1-methylimidazole (1.4 ml, 17.56 mmol) was added in a round bottomed flask fitted with a reflux condenser and the mixture was allowed to react at 358 K under stirring and N₂ atmosphere for 48 hr. The reaction mixture was washed twice with diethyl ether (30 ml) and then dried under vacuum at room temperature to result in a viscous orange/brown liquid [smmim][Cl[−]]. Yield: 3.2 g, 12.7 mmol, 95%.

Ion exchange step:

The complex of 1-methyl-3-(1-trimethoxysilylmethyl) imidazolium chloride (3.2 g, 12.7 mmol) was dissolved in CH₃CN (4 ml) and treated with 1.1 equivalent of sodium hexafluorophosphate (2.3 g, 13.7 mmol) dissolved in CH₃CN (40 ml). The resulting slurry was stirred for 5 days at room temperature and under N₂ atmosphere. The precipitate (NaCl) was removed by centrifuging at 9000 rpm for 20 min, while volatiles were removed by rotor evaporation at reduced pressure to give a yellow liquid [smmim][PF₆[−]].

2.2. Instrumentation

Permeability and selectivity evaluation were performed in a flow apparatus (Fig. 2) in which the feed, permeate and retentate gas compositions were analysed using a SRI gas chromatograph equipped with a stream selection valve, a gas sampling valve and TCD, FID detectors in series. Chromatographic separation was achieved with an Alltech Hayesep D 100/120 packed column (1/8", 30 ft). The gas flow of each stream was controlled by electronic mass flow controllers (Bronkhorst F-200CV) with a scale of 2–100 ml/min for CO (99.997%) and CO₂ (99.998%) and 20–1000 ml/min for He (99.999%) that was used as the sweep gas from the permeate side of the membrane. The pressure at both sides of the membrane was controlled by means of two electronic backpressure regulators (Bronkhorst P-702CV). The temperature was controlled with a PID controller and a Type K thermocouple in contact with the surface of the testing cell. All tests were performed in the Wicke–Kallenbach configuration with a mixture of 50/50 vol CO₂/CO sweeping the inner side of the tubular membrane (feed) and He sweeping its outer side (permeate).

The CO₂, CO gas solubility measurements were performed with a gravimetric microbalance (IGA, Hiden Analytical). The masses of the sample and counterweight pans, the hooks, the counterweight material and the hang chains of the microbalance assembly were of the order of one to three hundreds of milligrams per item and were defined with an accuracy of ±0.1%. The materials were appropriately selected to induce a symmetrical configuration to the balance set-up in order to minimize buoyancy effects. The microbalance had a 0.1 µg stable resolution. The amount of the bulk ionic liquid usually applied was 160 mg. Before each measurement the bulk ionic liquid samples were degassed at 353 K and high vacuum (10^{−5} mbar). The temperature dependent densities used for [bmim][PF₆[−]] in the buoyancy calculations were found from a linear fit of the data reported by Gu and Brennecke [27]. The densities of the bulk phase gas were calculated using the Benedict–Webb–Rubin equation of state for CO and CO₂. The uncertainty for this equation lies in on the order of 3% [28].

Elemental analysis (C, H, N) was performed with a Perkin Elmer 2400 CHN Analyser.

The cross sectional elemental composition of the membranes was analysed with a Field emission SEM (JEOL JSM-7401F) equipped with EDS analyser.

Thermal studies were conducted on a Setaram SETSYS Evolution 16/18, TGA/DSC analyzer.

2.3. SILM fabrication

2.3.1. Membrane activation-TGA

The effect of basic and acidic treatment on the surface hydroxyls population was investigated by means of TGA analysis in argon atmosphere. The α_{OH} values (µmol OH/g) presented in Table 2 were

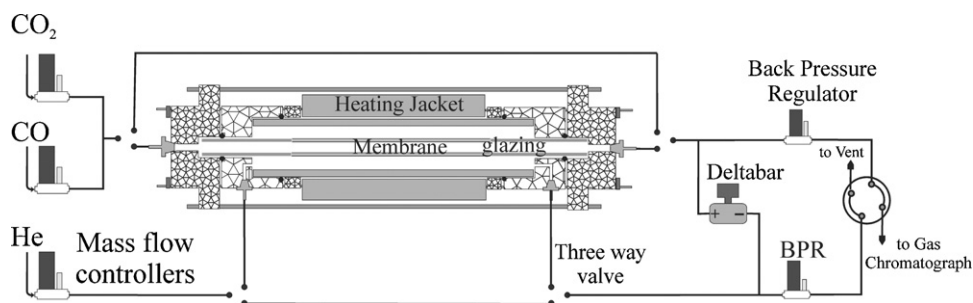


Fig. 2. The Wicke–Kallenbach apparatus applied for the measurement of the CO₂/CO selectivity.

Table 2

The surface activation conditions and the resulting hydroxyls concentration.

Entire membrane tube	
Conditions of treatment	TGA $\mu\text{molOH/g}$
Untreated membrane	17
H ₂ O ₂ /H ₂ SO ₄ (1/2), 20 min 60 °C	39
H ₂ O ₂ /H ₂ SO ₄ (1/4), 20 min 60 °C	61
NH ₄ OH PH 9.8, 6 h 25 °C	22
NH ₄ OH PH 9.8, 18 h 25 °C	15
NH ₄ OH PH 9.8, 20 min 90 °C	34
NH ₄ OH PH 9.8, 10 min 90 °C	48
NH ₄ OH PH 9.8, 1 h 60 °C	33
NH ₄ OH PH 9.8, 20 min 60 °C	71
NH ₄ OH PH 9.8 10 min 60 °C	41
<i>Silica nanofiltration layer</i>	
NH ₄ OH PH 9.8, 20 min 60 °C	150

calculated from the mass loss during the temperature ramp from 250 °C to 900 °C and the isothermal step at 900 °C. Before the ramp, an isothermal step at 250 °C was applied to completely remove the physically adsorbed water. The results show that under mild basic hydrolysis conditions (pH=9.8) and at each of the examined solution temperatures, there is a maximum of the α_{OH} value with the time of treatment. For treatment periods well in excess of the point where the maximum occurs, extended leaching of the surface leads to a moderate population of the surface hydroxyl groups. Acidic hydrolysis in a solution of H₂SO₄/H₂O₂, 4/1, for 20 min, presented moderate rehydroxylation efficiency compared to the NH₄OH solution (pH 9.8).

What is most important to note is the inhomogeneity in the population of surface hydroxyls along the asymmetric structure of the membrane. Indeed, after activation, the silica nanofiltration layer exhibited a double OH population compared to the entire membrane.

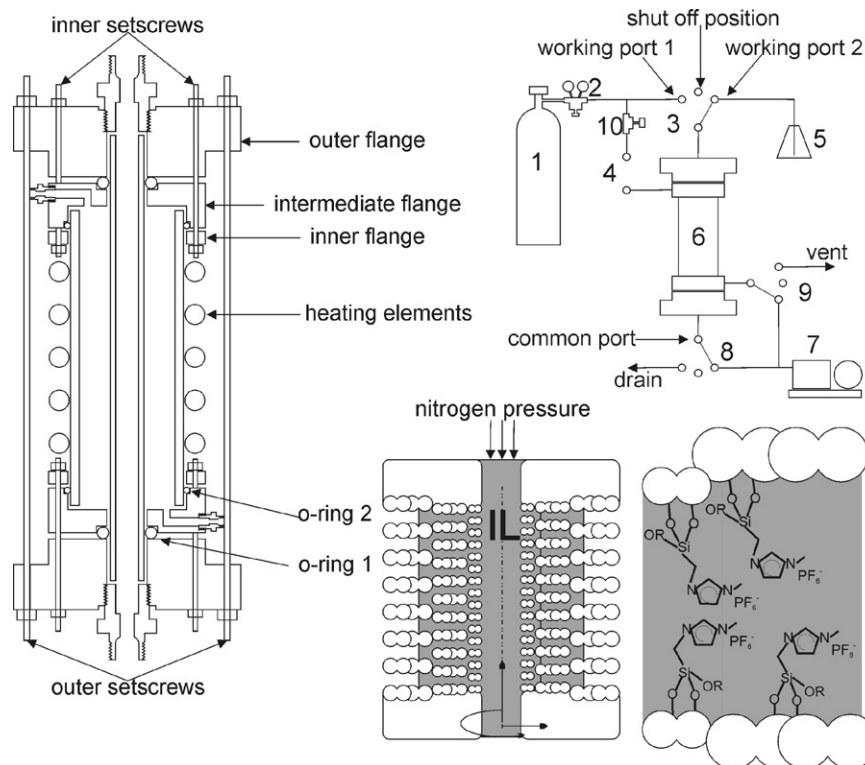
2.3.2. Membrane accommodation

After surface activation at the optimum conditions (NH₄OH pH 9.8, 20 min 60 °C), the tubular NF membranes were incorporated in a specially manufactured high-pressure stainless-steel reactor cell (Fig. 3), which apart from its use in the membrane modification procedure was further applied as the permeation cell for investigating the CO₂/CO separation performance. In this way we avoided contact of the SILM with the atmospheric air moisture. As it can be seen in Fig. 3 (left), the glazed ends of the tubular ceramic membrane (1.5 cm each) are initially passed through the borehole of two flanges (intermediate), while the inner flanges hold in place the tubular borosilicate glass cell of the reactor. Thereafter the outer flanges are introduced and squeezed together with the other two, by means of four internal setscrews. In this way the small diameter o-ring that embraces the glazed ends of the membrane, ensures efficient gas tight sealing between the outer and inner space of the membrane, whereas the large diameter o-ring, placed around the borosilicate glass cell, ensures the proper sealing between the outer space of the membrane and the environment. The outer setscrews allow for the involvement of elevated pressures in the internal space of the membrane tube and are tightened with a lower torque than this applied for the inner ones.

2.3.3. Membrane modification

2.3.3.1. Physical imbibition. The reactor cell is transferred to the imbibition/reaction device (Fig. 3 right) where the membrane is evacuated at both its sides for several hours by switching the three-way valves 8 and 9 to the vacuum pump (7) working ports and the valves 3 and 4 to the shut-off position.

The maximum temperature applied at this stage is 180 °C to ensure the preservation of most of the surface hydroxyls of the membrane. Thereafter, the temperature is decreased and a solution (0.6 M) of the silylated IL in ACN is suctioned under vacuum to the internal space of the membrane tube (lumen) by switching valve

**Fig. 3.** The complete set-up applied for the imbibition reaction modification of the tubular ceramic membranes.

3 to the IL solution (5) working port and valve 8 to the shut-off position.

As soon as the bore space of the membrane fills up with the solution, valve 3 is switched to the nitrogen flask (1) working port and the pressure is increased by means of the front pressure regulator (2). The pressure is applied to assure proper wetting of the pore structure and its maximum value depended on the pore size of the involved nanofiltration membrane. Then the high vacuum on the external side of the membrane is replaced by a gentle flow of nitrogen gas, generated by means of the second front pressure regulator (10) after switching valve 4 to the on position. The nitrogen stream sweeps the external surface of the membrane and flows out of the cell, through valve 9, which is switched to the vent working port. This configuration ensures sufficient pressure relief in case of membrane failure and, at the same time, the atmosphere at the external side of the membrane is inert during imbibition and subsequent reaction.

2.3.3.2. Reaction. The imbibition procedure ceases as soon as there is an optical indication for complete wetting of the membrane external area. This was achieved after about 1 h at the pressures of 7 and 1 MPa for the NF membranes with pores of 1 and 5 nm, respectively. At this point, the internal pressure is lowered to permit a very small flow rate of the IL solution through the pore structure. The temperature is increased to 80 °C and the reaction procedure starts and continues (see Fig. 3) until the collection of a 5 ml volume of the IL solution from the vent working port of valve 9. This corresponds exactly to the internal volume of the membrane tube, from the lower end of its upper glazed part to the common port of valve 3.

2.3.3.3. Washing and regeneration. After the completion of the reaction stage, the pressure in the internal side of the membrane is relieved down to 2 bars and the valve 8 is energized to the drain working port in order to remove the remaining IL solution. Then both the internal and external sides of the membrane are evacuated and pure solvent (ACN) is suctioned in the internal side of the membrane. The sequence of the solvent suction and draining is repeated for a minimum of three times and, finally, acetonitrile is driven to flow through the pores of the membrane under elevated pressure with a procedure similar to the one already described in Section 2.3.3.1. This stage is necessary to remove excess (unreacted) ionic liquid phase from the interior of the pores. Then the pressure is relieved; the solvent is drained and residual solvent is removed with the application of high vacuum on both sides of the membrane under elevated temperature (180 °C). Energizing all the surrounding valves to the shut-off position seals the reactor cell, which is then transferred to the permeability apparatus.

3. Results and discussion

3.1. Pore structure characterization of the NF layer

With the purpose to characterize the pore structure of the membranes' nanofiltration layer we conducted He permeability and LN₂ (77 K) porosimetry experiments. The permeability results for three silica membranes of the same batch are presented in the following Table 3.

The LN₂ porosimetry was performed on the bare alumina support and the entire silica membrane; both materials were provided by the ceramic manufacturer (Inopor®). The support isotherm was subtracted from that for the entire membrane and the derived isotherm was appropriately interpreted in order to calculate the open microporosity (ε) and pore surface area (A_c) of the silica nanofiltration layer. The raw isotherms and the one produced from

Table 3

Experimental values of He permeability and the calculated structure factors.

	Temperature (°C)		
	100	200	300
	He permeability (cm ² /s)/structure factor (κ_g)		
M1A	2.17E-06/6.4E-03	4.36E-06/1E-02	1.12E-05/5.9E-03
M1B	1.10E-06/3.3E-03	2.28E-06/6.0E-03	6.98E-05/3.7E-03
M1C	1.37E-06/4.0E-03	3.69E-06/9.7E-03	1.09E-05/5.8E-03

the subtraction procedure are presented in Fig. 4. From the geometrical dimensions and the solid density (d) of the materials consisting the NF layer (silica, $d = 2.2$ g/cc, thickness 1.5 μ m) and the support (alumina, $d = 3.2$ g/cc, ID = 7 mm, OD = 10 mm) we were able to calculate the mass ratio of the NF layer over the entire membrane. The following equation describes the isotherms subtraction procedure:

$$V_{NF} \left(\frac{cc(STP)}{g} \right) = \frac{V_{mem}(cc(STP)) - V_{sup}(cc(STP/g) \times M_{mem}(g) \times R_{sup}}{M_{mem}(g) \times R_{NF}} \quad (1)$$

where V_{NF} , the standard gas volume per mass adsorbed by the nanofiltration layer; V_{mem} the absolute standard gas volume adsorbed by the entire membrane; V_{sup} , the standard gas volume per mass adsorbed by the support; M_{mem} , the mass of the entire membrane; R_{sup} , the mass ratio of the support; and R_{NF} , the mass ratio of the nanofiltration layer. The derived NF layer isotherm was interpreted with the DR approximation to define the micropore volume and microporosity ($V_m = 0.128$ cc/g, $\varepsilon = 0.22$) and the micropore surface area ($A_c = 360$ m²/g).

The porosity ε and pore surface area A_c , were introduced to the following equation which gives the permeability factor D (cm²/s) according to the Fick's first law:

$$\frac{Jl}{U_c(\Delta C)} = D^0 \kappa_g = D = \frac{8}{3} \kappa_g \frac{\varepsilon}{A_c} \left(\frac{2RT}{\pi M} \right)^{\frac{1}{2}}, \quad \kappa_g = \frac{\delta \varepsilon}{\tau} \quad (2)$$

where D^0 (cm²/s), the thermodynamic or 'ideal' permeability factor for the case of parallel cylindrical pores of uniform size; κ_g , the structure factor; R , the gas constant; δ , depended on pressure, for low pressures $P_{\mu} \rightarrow 0$, $\delta \rightarrow (2-f)/f$, with f , the fraction of molecules striking the pore walls; M , the gas molecular weight; and τ , the tortuosity factor. The κ_g values for the three membranes were derived by the ratio between the experimentally defined and the ideal permeability factor and are included in Table 3. These values indicate a quite tortuous structure of the NF layer with the tortuosity factor (Eq. (2)) ranging between 20 and 60.

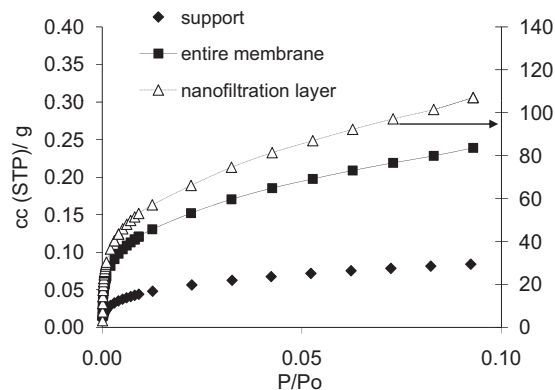


Fig. 4. The low-pressure area of the LN₂ isotherms performed on the support and the entire membrane.

Table 4
Elemental analysis results.

Silica nanofiltration layer					
C%	H%	N%	C/N	C/H	N/H
4.42	0.67	1.74	2.5	6.6	2.6
a-Alumina macroporous support					
C%	H%	N%	C/N	C/H	N/H
0.84	0.16	0.33	2.55	5.3	2.1
Predicted C%/N% ratios depending on the number of the grafted alkoxy groups of the silylated IL					
$C_7H_{14}N_2SiO_2PF_6$	3.0	6.0	2.0		
$C_6H_{11}N_2SiO_2PF_6$	2.6	6.5	2.5		
$C_5H_8N_2SiO_2PF_6$	2.1	7.5	3.5		

3.2. Elemental analysis/EDAX analysis

The following Table 4 presents the C, H, N content of the nanofiltration layer and the macroporous support of the 1 nm membrane (SILM1A), which was modified with the 1-methyl-3-(1-trimethoxysilylmethyl) imidazolium hexafluorophosphate. These sections of the tubular membrane are easily distinguishable and were mechanically removed and subjected to elemental analysis after the completion of permeability tests. Before the elemental analysis and in order to avoid the contribution of absorbed CO and CO₂, the membrane was out gassed at high temperature (180 °C) in the imbibition/reaction device.

A strong gradient of the surface concentration of the grafted ionic liquid (C%, N%) is found along the thickness of the membrane (see also Fig. 5); the simplest explanation is that the concentration gradient results completely from the already discussed (2.4.1) inhomogeneity in surface activity. According to the latter interpretation, at the initial stages of the imbibition/reaction procedure, the silylated ionic liquid molecules are mainly grafted on the surface of the nanofiltration pores. As a consequence, the concentration of the solution reaching the successive layers of the membrane must decrease continuously. Thereinafter as the reaction proceeds, the surface hydroxyls of the sequential layers, starting from the nanofiltration, are saturated and the silylated IL molecules are grafted on the surface hydroxyls of the next layer. Upon completion of the imbibition/reaction, the total volume of the solution (5 ml), having passed through the entire pore structure, contains an excess amount of the silylated IL molecules (3 mmol); the latter amount suffices for the saturation all of the surface hydroxyls of the membrane with an absolute population of 1 mmol as calculated by the net mass of the membrane tube (14 g) and the TGA-derived, OH-concentration (71 μmol/g, Table 2). Nevertheless, the α_{OH} ratio between the silica

NF layer and the entire membrane ~2.1 (Table 2), was much lower than the relevant ratio of the carbon concentration ~5.3 (Table 4); consequently the (-OH) gradient along the asymmetric membrane structure was not the only reason for the observed grafting inhomogeneity. In this context we might conclude that the anchoring of the silylated ionic liquid into the nanopores of the NF layer effectively hinders the passage of additional molecules into the pore structure of the succeeding layers. The latter interpretation is supported by Fig. 5, where we present the P% weight content defined by the EDAX analysis of the membrane cross section as a function of the distance from the NF layer. The x-coordinate corresponds to the center of the scanned area, with a width and a height of 40 μm. Thus, the first point pertains to the NF layer together with the main part of the subsequent g-alumina layer (see Fig. 1) and each of the remaining points is obtained by successive shifts of the x-coordinate by 5 microns in the direction from the inner to the outer side of the membrane. The horizontal lines represent the average C% and P% (w/w) contents, defined by scanning the top surface of the silica nanofiltration layer. The very sharp drop of the P% occurring up to a distance of 60 microns from the internal side demonstrates the sieving action of the grafted nanofiltration pores. On the other hand, the almost identical P% contents of the NF layer's top surface and cross section indicate that most of the excess silylated IL layer, which may have been formed due to the sieving mechanism on the top of the internal membrane's surface, have been effectively removed by means of the washing procedure described in Section 2.3.3.3. In addition, the C%/P% (w/w) content ratio for the top surface is about 2.4 instead of 3.1, which is the value expected from the stoichiometry of a non grafted silylated IL molecule (C₈H₁₇N₂SiO₃PF₆).

Moreover, both the C%/P% ratio of the top surface and the C%/N% ratios presented in Table 4 are compatible with the idea that grafting occurred with two of the methoxy groups of the silylated IL. This is evidence for the high density of the -OH groups being the result of the optimum activation conditions defined in this work. Finally although the derived by elemental analysis hydrogen content can be affected by the contribution of the surface hydroxyls, the good accordance of the C%/H% and N%/H% with those of the grafted silylated IL molecule (C₆H₁₁N₂SiO₂PF₆) is compatible with the idea that there is an almost complete consumption of the surface hydroxyls, especially those of the NF layer.

3.3. Absorption curves—CO₂, CO

In Fig. 6 we present the results of CO₂ and CO absorption at different temperatures for the ionic liquid [bmim][PF₆⁻] and its silylated analogue 1-methyl-3-(1-trimethoxysilylmethyl) imidazolium hexafluorophosphate (C₈H₁₇N₂SiO₃PF₆) [simmim][PF₆⁻]. As it can be observed, the carbon dioxide follows the exother-

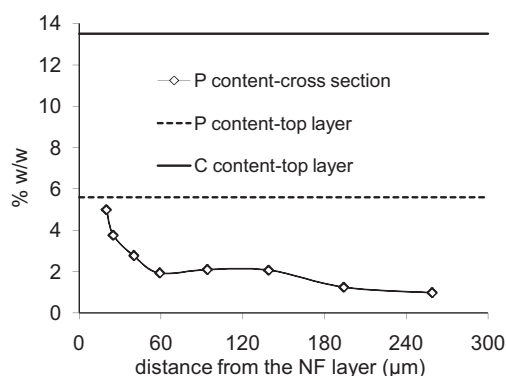


Fig. 5. EDAX elemental (C, P) analysis of the membranes' vertical cross-section and top nanofiltration layer.

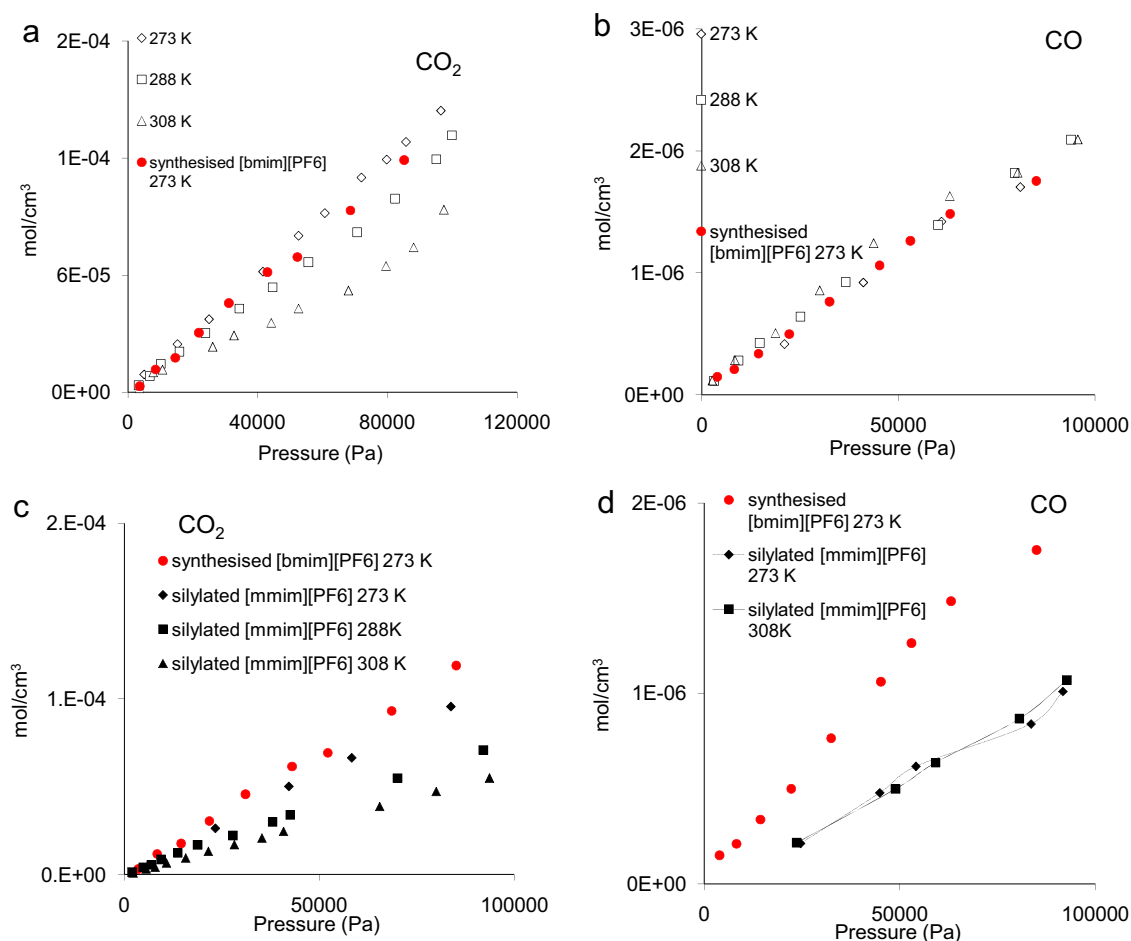


Fig. 6. (a) CO₂ absorption on the commercial and the synthesized bulk ionic liquid [bmim][PF₆], (b) CO absorption on the commercial and the synthesized bulk ionic liquid [bmim][PF₆], (c) CO₂ absorption on the synthesized silylated [mmim][PF₆], and (d) CO absorption on the synthesized silylated [mmim][PF₆].

mic character of solvation in both the commercial [bmim][PF₆[−]] and the [smmim][PF₆[−]]. The Henry constants ($\bar{H}$ bar/mol fraction) correlated positively with temperature. The $\bar{H}$ values of 32, 40 and 63 bar/(mol fraction) calculated for the commercial ionic liquid at 273, 288 and 308 K, respectively, were in good accordance with those often encountered in literature [9]. The $\bar{H}$ value of the synthesized [bmim][PF₆[−]] (33.8 bar/mol fraction) at 273 K, deviated no more than 6% from the value for the corresponding commercial product; this finding is compatible with the assumption that the synthesized ionic liquid phase is a substantially pure one. In the case of the synthesized alkoxysilyl-IL [smmim][PF₆[−]], absorption results were normalized per net mass of the 1-methyl-3-methylimidazolium-hexafluorophosphate [mmim][PF₆[−]] in the alkoxysilyl-IL molecule, since the silyl part is not expected to have any remarkable effect on the CO₂ absorption capacity.

By applying a (silyl/[mmim][PF₆[−]]) mass fraction of 0.5 as calculated from the stoichiometry of the molecule, we derived an $\bar{H}$ value of 42 bar/(mol fraction) at 273 K, 60 bar/(mol fraction) at 288 K and 80 bar/(mol fraction) at 308 K for the synthesized alkoxysilyl-IL. There are two possible reasons for the lower absorption capacity of the silylated-IL. The first has to do with the number of carbon atoms in the alkyl chain of the imidazolium ring, that is 1, for the [smmim][PF₆[−]] and 4, for the bulk ionic liquid [bmim][PF₆[−]]. Various studies suggest that apart from anionic part of the IL, changes in the imidazolium cation involving alkyl groups might also affect the CO₂ solubility [5]. In fact the longer alkyl chains usually lead to slightly better CO₂ absorption capacity in the ionic liquid phase

[29,30]. In our case the effect of the carbon atom number seems to be more enhanced as the Henry constant ratio, for a difference of 3 carbon atoms in the alkyl chain, was about 0.7 compared to ratios of 0.94 often found in literature for differences of 2 up to 4 carbon atoms [29]. Based on this fact we might conclude that the lower CO₂ solubility value mainly arises as a result of the presence of a silyl group, attached to the alkyl chain of the imidazolium ring; the latter feature might lead to a modest loss of organization of the anion and the CO₂ about the cation.

As regards the CO solubility (Fig. 6b and d), the calculated $\bar{H}$ values of 2158, 2185 and 2373 bar/(mol fraction) at 308, 288 and 273 K for the commercial [bmim][PF₆[−]] and 2202 at 273 K for the synthesized [bmim][PF₆[−]], were in good accordance with those found in the work of Kumelan et al. [31], and deviated positively by approximately 100% from those found in the studies of Jacquemin et al. [32]. Moreover in both of the aforementioned studies and also in our work the calculated $\bar{H}$ values for CO practically did not depend on temperature. As a matter of fact in the case of the work of Kumelan et al. [31] there was even an indication of a slightly negative correlation and the same is true in our case ($\bar{H}$ = 2373, 2185 and 2158 bar/(mol fraction) at 273, 288 and 308 K, respectively); such trends indicate that the crossed gas–ionic liquid molecular interactions are of different nature in the CO and CO₂ cases. In Table 5 we present the CO₂ and CO solubility values expressed in mol/cm³/Pa and the derived ideal sorption CO₂/CO selectivities. As it can be seen the silylation of the imidazolium based ionic liquid had a more intense negative effect on the solubility of CO compared to the CO₂, thus resulting in an enhanced CO₂/CO separation capacity.

Table 5
Solubility of CO₂ and CO and the derived ideal selectivities.

T (K)	308	288	273
	Solubility (<i>S_i</i>) 10 ¹⁰ × (mol/cm ³ /Pa) <i>S_{CO2}</i> / <i>S_{CO}</i> /(selectivity CO ₂ /CO)		
Commercial [bmim][PF ₆ [−]]	8.61/0.24/ 36	12.5/0.23/ 54	15/0.21/ 72
Silylated [smmim][PF ₆ [−]]	5.91/0.12/ 50	7.7/–/–	11.5/0.11/ 105

In general the selectivity values observed suggest that the potential of membrane CO₂ separation technology by means of a SILM system is a substantial one.

3.4. Transient curves—CO₂ diffusivity

For the definition of the diffusivity *D* (cm²/s) of CO₂ in the commercial ionic liquid and the silylated IL, the transient curves of the CO₂ adsorption at 308 K were fitted to the solution proposed by Crank for the adsorption in a plane sheet at the beginning of a sorption experiment when the following conditions are satisfied [33]: (1) the initial concentration of the gas in the liquid is uniform, (2) the surfaces of the liquid film are kept at constant concentration, and (3) the amount of the gas taken up by the liquid is a negligible fraction of the whole. The solution has the following form:

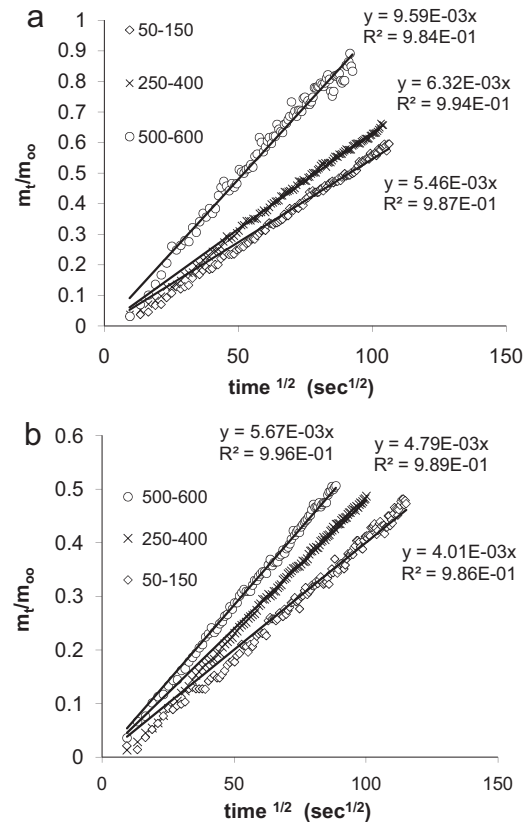
$$\frac{m_t}{m_\infty} = 2 \left(\frac{Dt}{\ell^2} \right)^{-\frac{1}{2}} \Leftrightarrow \left\{ \pi^{\frac{1}{2}} + 2 \sum_{n=1}^{\infty} (-1)^n \operatorname{erfc} \frac{n\ell}{\sqrt{Dt}} \right\} \quad (3)$$

where *m_t* the total amount of gas, which has diffused into the liquid film at time *t*; *m_∞* is the total amount of gas that has diffused into the liquid after infinite time and *ℓ* (cm) the half liquid film thickness. For very short times, the fractional uptake of the gas by the liquid is approximated well by the simple equation:

$$\frac{m_t}{m_\infty} = 2 \left(\frac{Dt}{\pi \ell^2} \right)^{-\frac{1}{2}} \Leftrightarrow \quad (4)$$

The diffusivities for the several pressure steps were calculated from the slope of the *m_t/m_∞* vs $\sqrt{t}$, for a liquid film thickness 2 *ℓ* of about 0.15 cm, as calculated by the liquid mass and density and the diameter of the sample basket used in the microbalance setup. The results are presented in Table 6.

The *m_t/m_∞* vs $\sqrt{t}$ plots for the synthesized ionic liquid and the silylated IL are depicted in the Fig. 7. In both examined cases the diffusivity increases with pressure as the result of the decrease of the ionic liquid viscosity due to the dissolution of higher amounts of CO₂.

**Fig. 7.** *m_t/m_∞* vs $\sqrt{t}$ plots at several pressure steps (a) [bmim][PF₆[−]] and, (b) [smmim][PF₆[−]].

The diffusivity constants (*D*) of 10^{−6} cm²/s coincide with the results of Hou and Baltus [34]. Finally, the predicted CO₂ permeability values are also included in Table 6. They were calculated

Table 6
Permeability factors calculated from the transient curves of the absorption experiments.

	Pressure steps (mbar)		
	50–150	250–400	500–600
	$2(D/(\pi \ell^2))^{1/2}$ (s ^{−1/2})/ <i>D</i> (cm ² /s)		
Commercial [bmim][PF ₆ [−]]	0.0055/1.7E-06	0.0063/2.33E-06	0.0096/5.4E-06
Silylated [smmim][PF ₆ [−]]	0.0041/9.8E-07	0.0048/1.35E-06	0.0057/1.91E-06
	Permeability (10 ¹⁶ × (mol cm/cm ² /s/Pa))/Barrer		
Commercial [bmim][PF ₆ [−]]	14.63/4368	20/5971	46.5/13883
Silylated [smmim][PF ₆ [−]]	5.8/1732	7.98/2383	11.28/3368

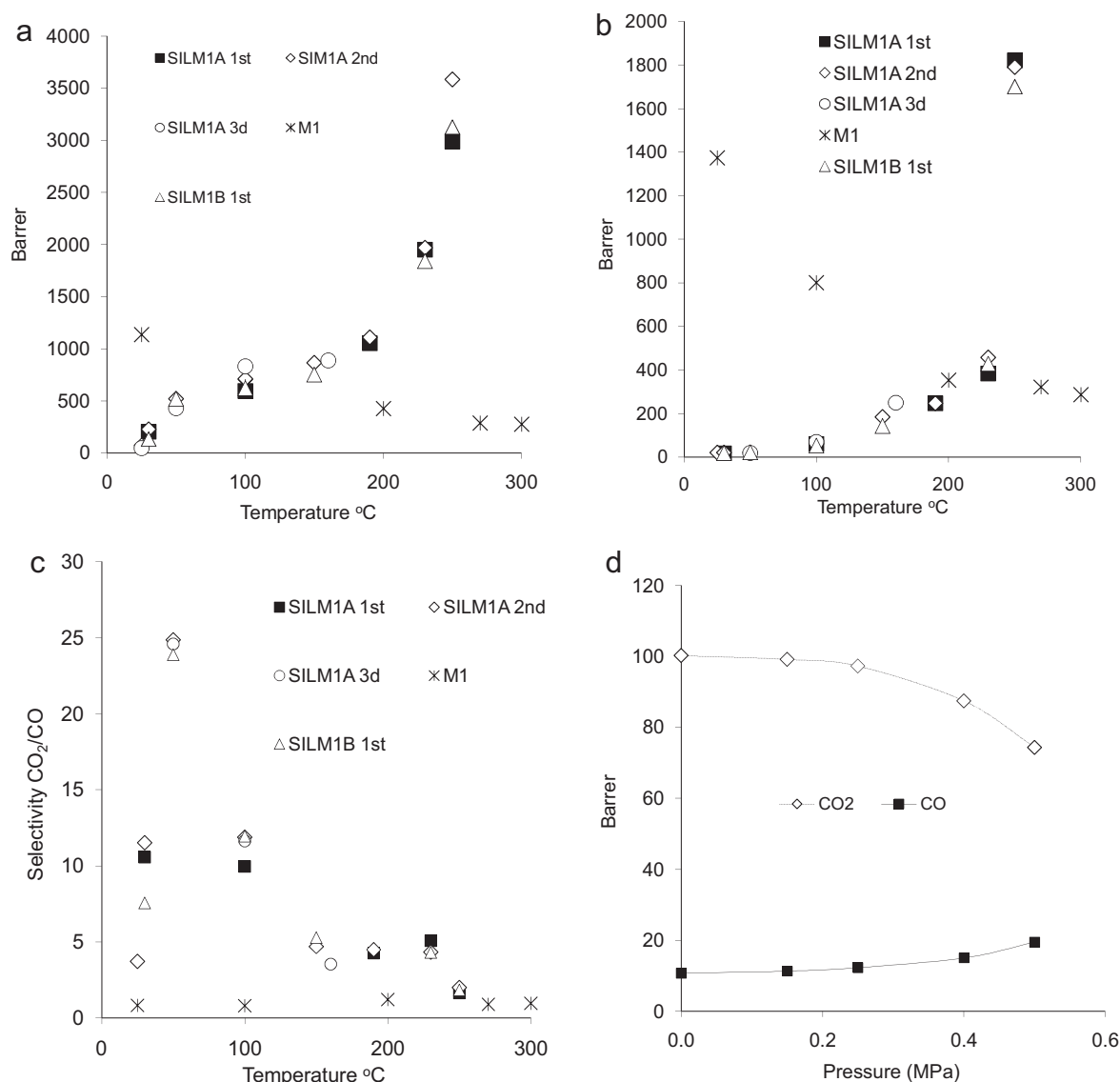


Fig. 8. (a) Permeability of CO₂, dependence on temperature, (b) permeability of CO, dependence on temperature, (c) CO₂/CO real selectivity, dependence on temperature, conditions for (a–c): trans-membrane pressure = 0.5 MPa, feed conc. = 50%/50%, CO₂/CO (v/v), (d) permeability dependence on pressure at 25 °C.

with the assumption that transport through a SILM follows the solution/diffusion mechanism:

$$P = DS \quad (5)$$

where P is permeability; S is solubility (in moles per volume per partial pressure), and D is diffusivity.

A remarkable observation is that the presence of the silyl group on the alkyl chain of the imidazolium ring, not only resulted to considerably lower CO₂ absorption capacity, but also decreased the rate of diffusion to about half the value of these of the commercial [bmim][PF₆[−]] sample.

3.5. Diffusion mechanism-separation performance

The stability of the developed SILM membranes and the reproducibility of the proposed imbibition/reaction method can be concluded from the permeability selectivity results presented in Fig. 8. The duration of testing at each temperature was about 4 h and, for example, membrane SILM1A which was subjected to three sequential heating/cooling cycles from R.T to 250 °C, at a trans-membrane pressure of 0.5 MPa, exhibited a 5-day sta-

bility. In addition, the similarly developed membrane SILM1B exhibited permeability factors within $\pm 10\%$ of those defined for SILM1A.

The permeability was calculated with the assumption that the pore volume of the thin nanofiltration layer (1.5 μm thickness) is completely occupied by the silylated ionic liquid phase [smmim][PF₆[−]], which primarily controls the gas diffusion through the entire SILM. Indeed, the CO permeability of the pristine nanofiltration membrane M1 at room temperature is about 100 times higher than this of the modified SILM membranes; this finding is indicative of the high extent of pore blockage. An issue that requires further explanation is that at the temperature of 30 °C, the experimentally derived permeability values of CO₂ for the modified membranes SILM1A and SILM1B (Fig. 8a) are about one order of magnitude lower than those calculated by means of Eq. (5), (Table 6), with the factors D and S already defined from the CO₂ isotherm of the silylated ionic liquid [smmim][PF₆[−]] (Table 5).

This enhanced differentiation must be seriously considered when trying to determine the upper limits, benchmarks, and critical properties for gas separations using SILMs [35], solely with the help of predictive correlations [34,36–43] for gas solubilities and

Table 7

Experimental permeability values measured with the Wicke–Kallenbach set-up.

		<i>T</i> °C	Pe CO ₂ Barrer	Pe CO Barrer	Selectivity CO ₂ /CO
M5	–	25100	2741227365	2849627941	0.960.98
SILM5A	C ₈ H ₁₇ N ₂ SiO ₃ PF ₆	25100	2298723124	2737527564	0.840.84
SILM5B	C ₁₀ H ₂₁ N ₂ SiO ₃ PF ₆	25100	454460	624672	0.730.68
M1	–	25100	1135635	1373800	0.830.8
SILM1A	C ₈ H ₁₇ N ₂ SiO ₃ PF ₆	25100	226715	2060	11.311.9
SILM1C	C ₁₀ H ₂₁ N ₂ SiO ₃ PF ₆	25100	12484	15479	0.811.07

diffusivities in RTILs. Several factors may contribute to the differentiation from the predictions, each one with a different impact, such as:

- The counter diffusion of the sweep gas applied in the permeability–selectivity tests.
- The fact that even if the pores are completely filled with the ionic liquid, the IL permeability is underestimated due to the presence of the non permeable ceramic in the remaining fraction of the membrane volume (porosity) and the tortuosity of the pore structure.
- The presence of the second type of gas molecules that dissolve simultaneously with the CO₂ molecules in the bulk of the silylated ionic liquid.
- The contribution of the silylated ionic liquid phase that is deposited on the pore surface of the subsequent macroporous membrane layers, without blocking, at the same time, the pore space. This contribution is indirectly demonstrated on the basis of the data presented in Table 7. From the comparison of the cases of M5 (non grafted nanoporous) and SILM5B (grafted but non blocked nanoporous) it is clear that the CO₂/CO selectivity is worse in the latter case, i.e. when the grafted material does not block the pores it favors the passage of CO over that of CO₂.
- The application of elevated trans-membrane pressures (0.5 MPa).

The negative effect of the first two factors has already been discussed in the literature [12,43–45]; the larger of the two contributions corresponds to the second factor and depends on the porosity ε of the membrane and the tortuosity factor τ (usually $\varepsilon = 70\%$ and $\tau = 1.2$ – 2 , for macroporous polymeric membranes). The ceramic nanofiltration membranes used in this work had a moderate open microporosity of 22%, whereas the tortuosity (τ) exceeded the value of 20 (see Section 3.1). According to these pore structural characteristics the experimentally defined CO₂ permeability of the ionic liquid should be 20/0.22~100 times lower than the predicted one. Thereby, we can claim that the ionic liquid permeability is overestimated rather than underestimated. A possible explanation for this discrepancy is that the routes of the permeating CO₂ molecules inside the pores are less tortuous than those in the bulk liquid because the grafted material inside the pores is a structured one (see also the discussion about the maximum in CO₂/CO selectivity as a function of temperature).

In what follows we provide evidence that the ionic liquid phase, which is deposited on the pore surface of the subsequent macroporous membrane layers, may have a considerable contribution to the reduction of the CO₂/CO separation performance. To this end we present (Table 7) the results obtained for the modified gamma-alumina NF membrane (SILM5) with nanofiltration pores of 5 nm. The latter size was too large for complete pore filling by the surface anchored [smmim][PF₆][−] species; Because of the openings left, the combined result was a CO₂/CO selectivity below one.

It can be concluded that when the deposition of the silylated IL was not accompanied by complete blockage of the nanopore struc-

ture (SILM5A), we had an opposite effect on the CO₂/CO separation capacity, e.g. the CO permeated faster.

Based on these findings, a complex diffusion mechanism is proposed for the [smmim][PF₆][−] modified nanofiltration membranes of 1 nm pore size (Fig. 9). Specifically, at the section of the nanofiltration layer (A), the preferable solvation of CO₂ into the silylated ionic liquid favors its separation from CO.

Thus more CO₂ molecules diffuse through the bulk ionic liquid and desorb into the gas phase inside the void space of the succeeding macroporous layers. Next to that, partitioning of the CO₂ and CO molecules occurs between the gas phase and the ionic liquid film. The partitioning mechanism favors the faster permeation of CO, since the CO₂ molecules reside for a longer time into the liquid phase. The partitioning effect must be strong, due to the minor diffusion length of the nanofiltration layer (1.5 μ m) compared to this of the succeeding macroporous layers (1500 μ m). However we cannot accept it as the only reason for the CO₂/CO selectivity decrease from 50, as predicted from the absorption results (Table 5), to about 11 as defined from the permeability experiments (Table 7). On the basis of the data presented in Fig. 8d, we suggest that the elevated trans-membrane pressure and the co-solvation/diffusion of CO have a negative effect on the permeability of the CO₂ species. The pressure plot demonstrates that for a 50/50% (v/v) mixture of CO₂/CO, the trans-membrane pressure increments have provoked considerable reduction of the CO₂ permeability and an exactly analogous increase of the CO permeation. Contrary to this, the analysis of the single-phase CO₂ absorption transient curves revealed a positive correlation of the CO₂ diffusivity with pressure, as expected due to the decrease of the IL viscosity upon absorption of higher amounts of CO₂.

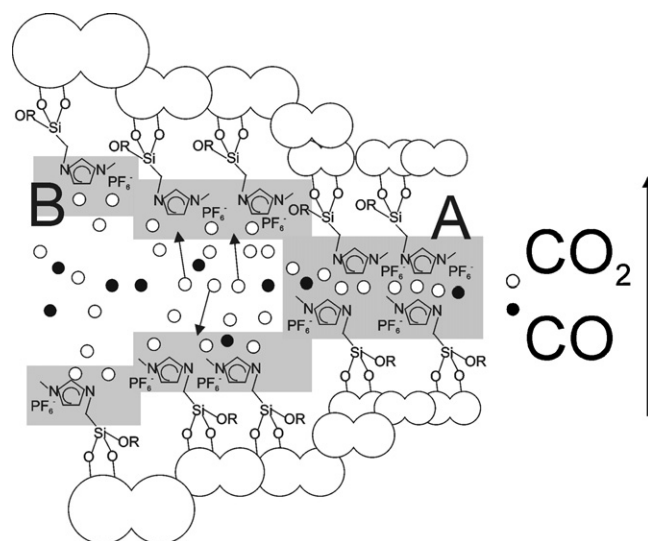


Fig. 9. The proposed overall diffusion mechanism showing the contribution of the uncovered macroporous layers of the asymmetric membrane.

In this context we can argue that the reason behind the negative correlation of the CO₂ permeability with pressure is ultimately due to the increasing amounts of CO molecules, which modify strongly the CO₂–ionic liquid interaction. In a study concerning the system ionic liquid/carbon monoxide [46], conducted by high pressure NMR C¹³ spectroscopy, it was shown that the CO solubility depends upon the nature of the ionic liquid due to the presence of dipole moment and polarisability. In the same study concerning the 1-butyl-3-methylimidazolium cation based ionic liquids, the solubility of CO increased with increasing anion size due to the decrease in π^* interactions. Thereby the interaction of the CO molecules with the anion may lead to the loss of the “tangent like” organization of the CO₂ molecules around the anion, which is regarded as the main reason for the high CO₂ solubility [5].

More generally the case of pairs of CO₂ with other non-absorbant gases such as CO, N₂, O₂, H₂, CH₄ requires further investigation. The effect of pressure on the selectivity of CO₂ in “real” gas mixtures is rarely investigated in literature and in a recent study [43] it was suggested that methane partial pressure did not significantly alter the CO₂ permeance or selectivity for the RTIL-membranes tested, although a slight decrease of the CO₂/CH₄ selectivity can be appreciated when examining the relevant plots. Additionally in another study [21] of the “ideal” CO₂ selectivity over CH₄, the authors suggested that the actual CO₂/CH₄ selectivity will not be influenced by pressure; on the basis of our CO₂/CO data it appears possible that the situation will be reversed also in the real case of a CO₂/CH₄ mixture.

Concerning the performance dependence on temperature, it is interesting to note that the CO₂/CO selectivity presented a maximum at 50 °C for both membranes SIM1A and SILM1B. This indicates that the solution-diffusion mechanism is not a simple one. Selectivity would normally be a monotonous function of temperature in a bulk material where local structural subtleties are averaged. On the other hand in our case the structure of the matrix material (grafted pore-filling material) exhibits a variation along the radius and at least in the case of CO₂ species there might be a preference for certain, far from random, paths that favour Lewis-acid based interactions. When the temperature is raised individual CO₂ diffusion steps might become easier but at the same time the intraporous material might become less ordered and hence the original path might become more tortuous/less efficient; the outcome of the combined affect of the latter two factors allows for the exhibition of subtleties (including non-monotonous variation of selectivity with temperature) which are not encountered in the usual case of bulk matrices. We intend to explore via further single-phase permeability experiments in various temperatures the details of actual processes.

We might also note the results reported for a non grafted SILM (supported ionic liquid membrane): a maximum in CO₂/H₂ selectivity has been observed for $T=95\text{ }^{\circ}\text{C}$ [15]; interactions and various other details (e.g. there is no grafting) differ from ours but it is interesting that a maximum in selectivity with temperature is again possible.

Finally from the results presented in Table 7 we can conclude that the longer alkyl chain of the silane molecule (propyl chloro, instead of methyl chloro silane) was not sufficient to provoke the complete blockage of the 5 nm pores (SILM5B). Although the permeability factors were considerably reduced compared to membrane SILM5A, the CO₂/CO selectivity was still inversed, thus confirming the partitioning effect, which occurs in cases where the ionic liquid phase is just deposited on the pore surface without effectively blocking the pore space. On the other hand, the propyl chain was long enough to prohibit the entrance of the silylated-IL (1-methyl-3-(3-trimethoxysilylpropyl) imidazolium hexafluorophosphate) into the 1 nm pores of membrane SILM1C. In this case, the permeability was about one order of magnitude

lower than this of the unmodified membrane as a consequence of the deposition of the silylated IL molecules on the external surface of the top-separating layer, around the pore mouths. Thus the pore mouth openings were partially blocked resulting to considerable hindrance on the entrance of the gas molecules into the pore structure but not to a considerable enhancement of the CO₂/CO selectivity since the deposited IL phase was not exerting full control over the diffusion of the gaseous molecules.

4. Conclusions

A novel pressure-assisted imbibition/reaction method is proposed, for the chemical stabilization of silane functionalized imidazolium based ionic liquids into the pores of ceramic nanofiltration membranes. Stabilization is for the first time achieved via chemical bonding between the solid surface hydroxyls and the alkoxy group of the silyl part of the functionalized ionic liquid molecule, rather than with the help of capillary forces and ionic interactions. The membranes exhibit considerable CO₂/CO separation capacity (25) and excellent stability at temperatures up to 250 °C. The method was proved efficient and reproducible in the case of membranes with 1 nm pore size and ionic liquids, functionalized with the chloromethyl trimethoxy silane. For larger pores (5 nm), alkyl chains of more than 3 carbon atoms are needed for effectively covering the entire pore space with the deposited silylated molecule. In this context further studies must be conducted in order to assess the possibility to cover nanofiltration pores of larger size and to define the optimum silylated analogues for pore sizes from 1 to 5 nm; we intend to explore the latter issues in a subsequent work. Although the silylated IL exhibited moderate CO₂ absorption capacity compared to the pristine [bmim][PF₆][−], the absorption separation capacity over CO was maintained or, even, was enhanced; hence the approach employing silanization and subsequent grafting of ionic liquids for the development of stable SILM systems deserves further attention.

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